



Smt. Chandaben Mohanbhai Patel

Institute of Computer Applications



The Real Time Group Expense Sharing

- Team Members •

MANAN KANSARA - 24MSIT041

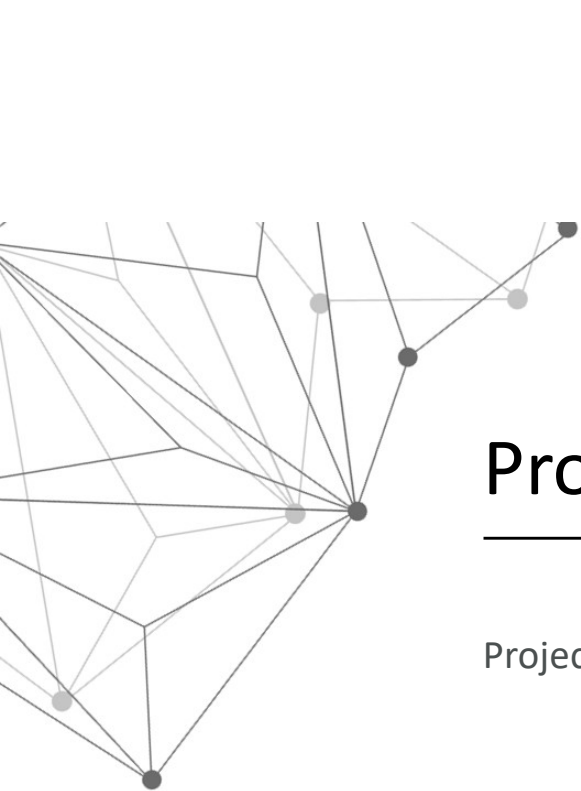
NIDHI PATEL - 24MSIT101

NEEL UDHANAWALA - 24MSIT182

- Guide's Name •

MR.DIP PATEL



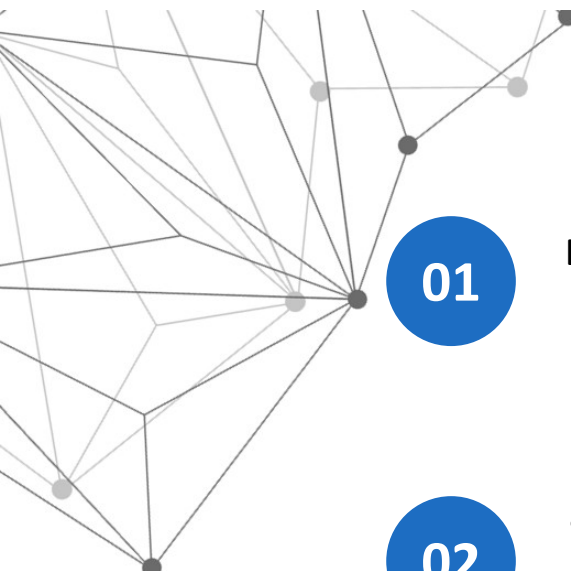


Project Profile & Comapny Profile

Project Description:

A web-based platform designed to split, track, and manage real-time expenses among groups and their members. The system allows users to create groups, add participants, log shared expenses, and automatically calculate each member's share. It supports real-time updates, payment tracking, and settlement summaries, ensuring transparent and efficient financial collaboration for activities like trips, events, or shared households.





Introduction to Tools

01

Development Tools

Visual Studio Code – A source-code editor used for writing, debugging, and managing the project

02

Technologies

Front-end Tools :React – A JavaScript library for building dynamic and interactive user interfaces.

Bootstrap – A framework for creating responsive, mobile-first web pages.

Back-end Tools : Node.js – A JavaScript runtime for server-side programming.

Express.js – A lightweight framework for building APIs and handling server logic.



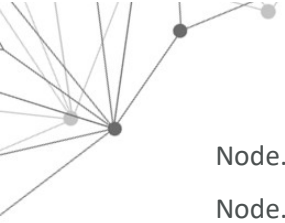


React:

React is a powerful open-source JavaScript library developed by Facebook for building modern, interactive user interfaces. It follows a component-based architecture, allowing developers to break down complex UIs into smaller, reusable parts. This approach improves maintainability, scalability, and efficiency. React uses a virtual DOM, which enhances performance by updating only the necessary elements on the page rather than reloading the entire interface. Its unidirectional data flow simplifies debugging and ensures predictable behavior across components. React also supports easy integration with other frameworks and libraries, making it ideal for building dynamic, high-performance web applications.

Bootstrap:

Bootstrap is a popular front-end framework that helps developers design responsive and mobile-first web applications quickly. It provides a wide range of pre-designed components, such as navigation bars, buttons, modals, and forms, which ensure consistency in design and layout. Bootstrap's powerful grid system allows the interface to automatically adapt to different screen sizes, providing an excellent user experience across devices. It also includes built-in support for modern browsers and offers extensive customization options through CSS and JavaScript. By using Bootstrap, developers can save time in UI design and focus more on core functionality while maintaining a clean and professional look.

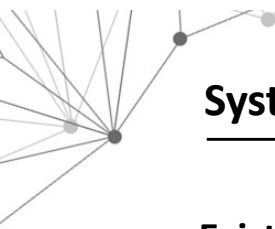


Node.js:

Node.js is an open-source, cross-platform runtime environment that allows developers to run JavaScript on the server side. Built on Google's V8 JavaScript engine, Node.js delivers exceptional performance and speed. It uses an asynchronous, event-driven architecture that enables non-blocking I/O operations, allowing the application to handle thousands of concurrent requests efficiently. This makes Node.js ideal for building scalable, real-time, and data-intensive web applications. Since it uses JavaScript, developers can maintain a unified language across both the client and server sides, improving development speed and consistency. Node.js also has a large ecosystem of open-source libraries available through npm, which accelerates the development process.

Express.js:

Express.js is a minimal and flexible web application framework built on top of Node.js. It provides a robust set of features for building APIs and web applications efficiently. Express simplifies the process of defining routes, handling requests and responses, and managing middleware. It supports RESTful API design, making communication between the front-end and back-end seamless. The framework also allows easy integration with databases and third-party services, ensuring a smooth data flow. Express.js promotes cleaner code organization and reduces development time, making it one of the most widely used back-end frameworks in modern web development. It forms the backbone of the server-side logic in the application.



System Study(Requirement Analysis)

Existing System

At present, people usually manage shared expenses in two main ways:

Manual methods such as writing amounts on paper or using spreadsheets.

Basic mobile apps that don't always offer real-time synchronization or clear visibility for everyone.

Both approaches can easily lead to mistakes, slow updates, and even arguments among group members.

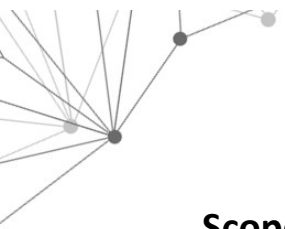
Proposed System

SplitBuddy is designed to solve these problems by offering a single, easy-to-use platform where groups can manage all their shared expenses. The system provides:

A centralized space to record and monitor expenses.

Instant updates so every member sees changes as soon as they happen.

Secure settlement tracking and clear reports to keep everyone on the same page.



Scope of the Proposed System

The application is suitable for any situation where people share costs, including:

Trips, tours, and vacations.

Roommates or shared households.

Group events, office parties, or club activities.

Any other collaborative activity that needs transparent expense management.

Aims and Objectives

SplitBuddy's main purpose is to make group expense sharing simple, fair, and transparent. Its objectives are to:

Automate how expenses are divided among group members.

Keep a clear history of settlements and provide detailed reports

Enable safe, real-time collaboration on any device with an internet connection.



Feasibility Study

Operational Feasibility

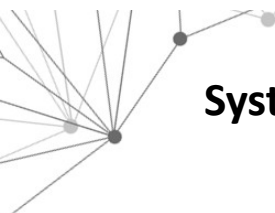
The application is web-based with a clean, intuitive interface. Users need little to no training to start managing their expenses.

Technical Feasibility

SplitBuddy is built using proven technologies—React for the front end, Node.js with Express for the back end, and MySQL for data storage. These tools are widely supported, scalable, and reliable.

Economical Feasibility

Development and upkeep costs are relatively low, and the app can be deployed on affordable cloud hosting services, making it budget-friendly.



System Analysis

Requirements Specification (With System Modules)

Functional Requirements

- User registration and authentication
- Group creation, member management
- Add, edit, delete expenses
- Automatic calculation of each member's share
- Settlement management
- Notifications
- Admin support (manage user , group monitoring , resolve queries)

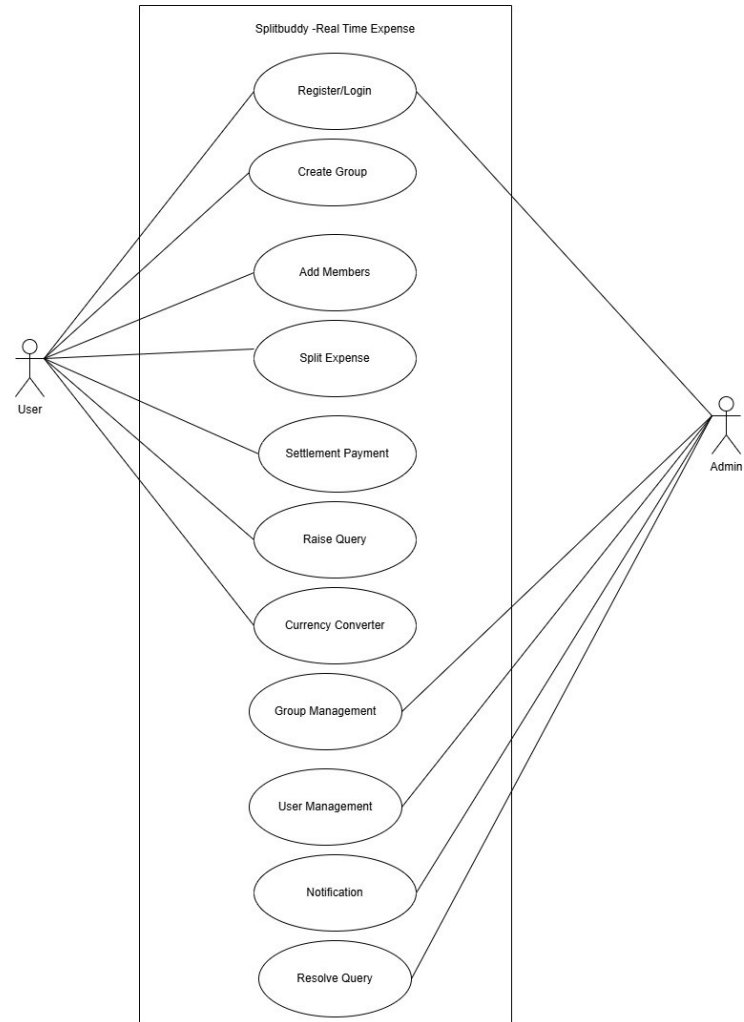
Non-Functional Requirements

- Responsive UI
- Data security (password hashing, role-based access)
- High availability and scalability
- Real-time updates via APIs/web sockets

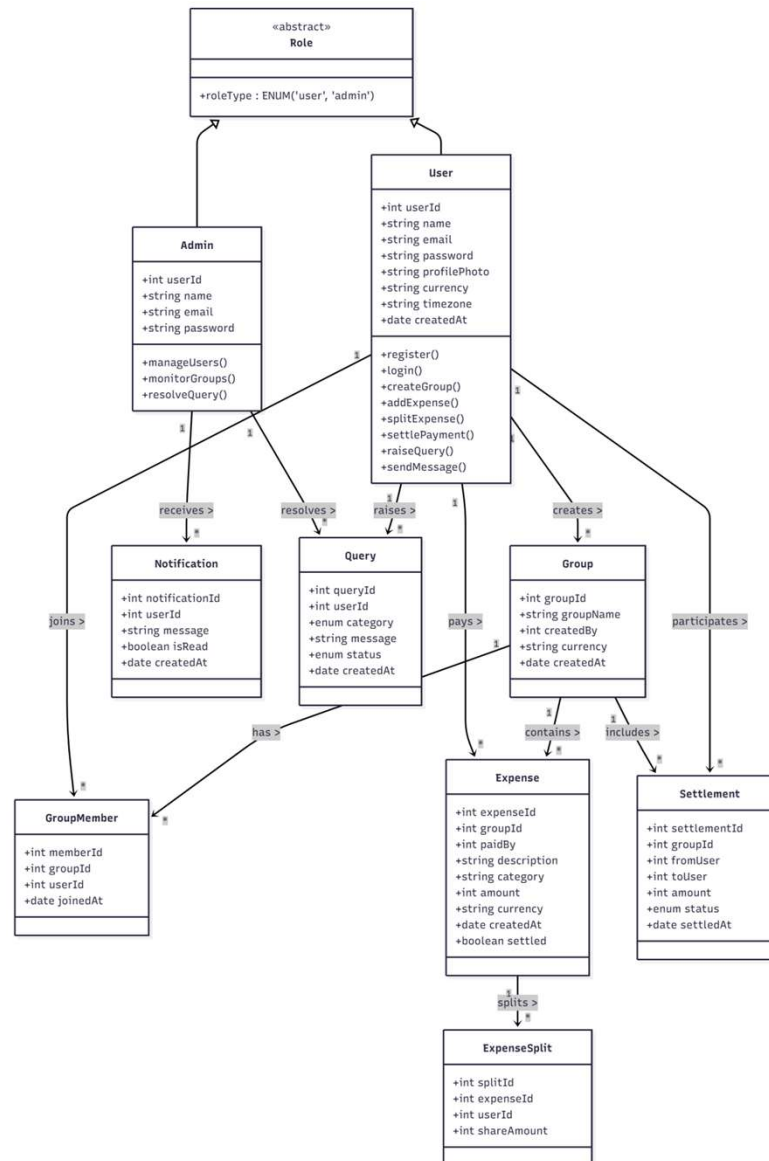
System Modules

- User Module
- Group & Member Module
- Expense & Split Module
- Settlement Module
- Notification Module
- Admin Support Module

Use Case Diagram

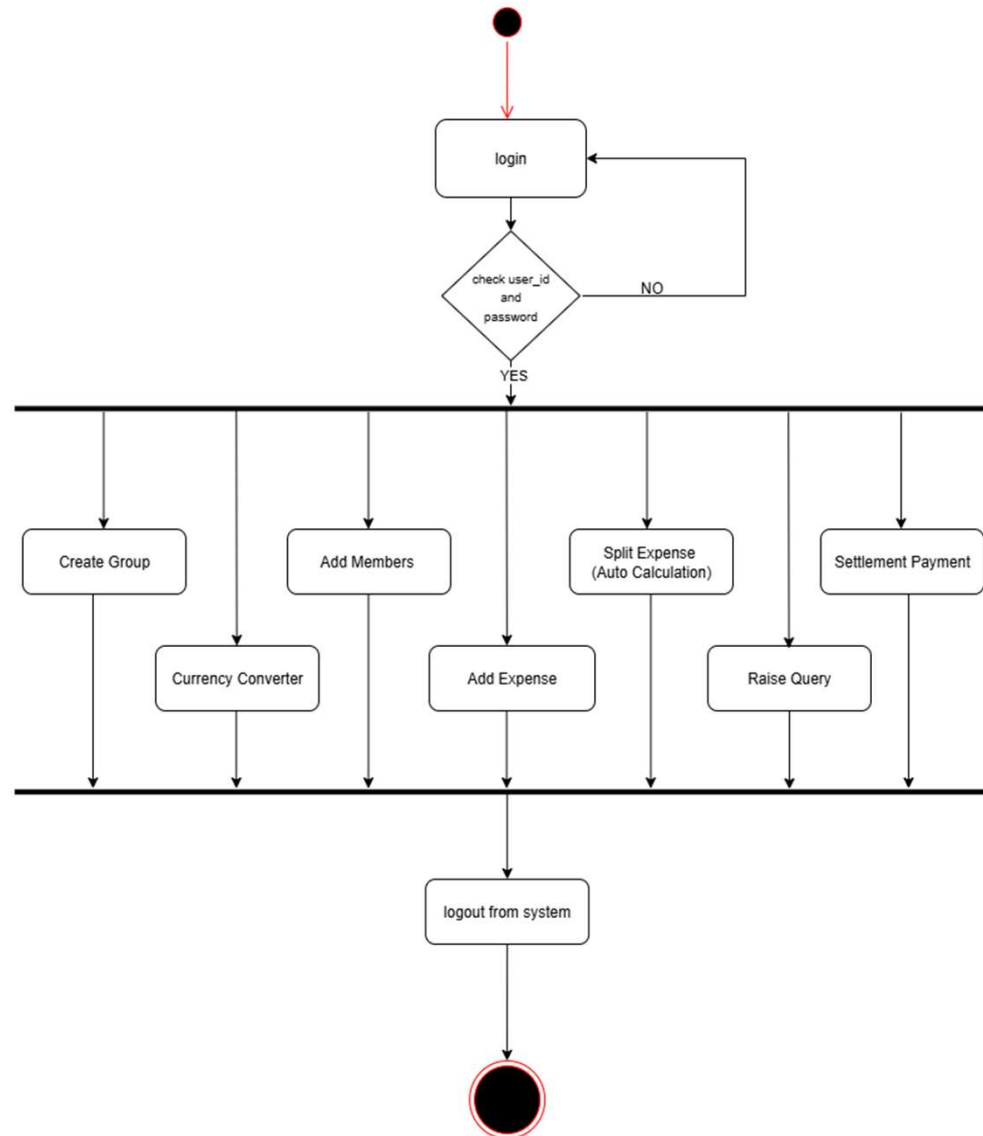


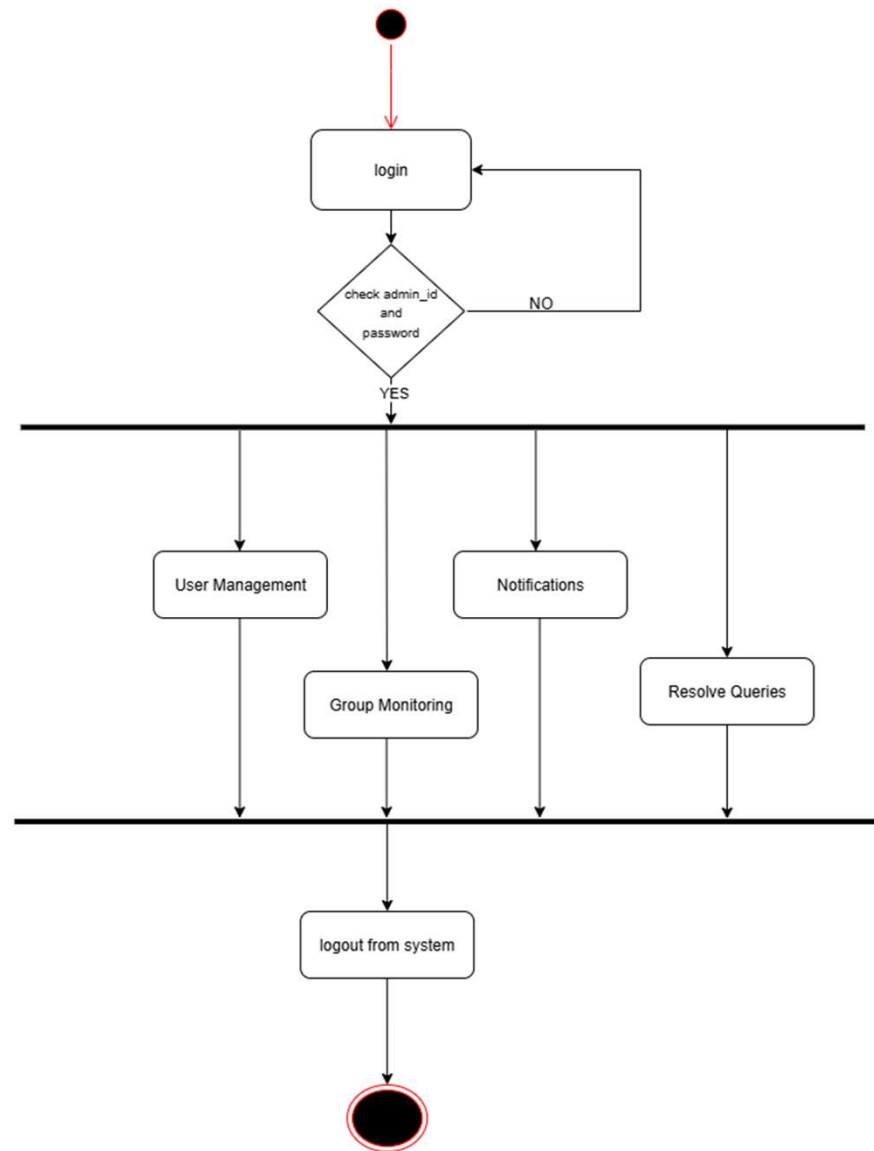
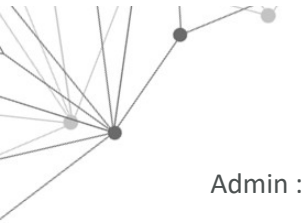
Class Diagram / System Flowchart

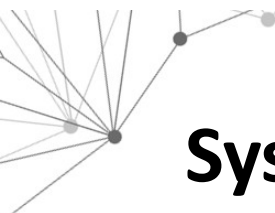


Activity Diagram

User :







System Design

Data Dictionary

Table Name: users

Column Name	Data Type	Constraints / Default	Description
user_id	INT (AUTO_INCREMENT)	Primary Key , Auto-increment	Unique identifier for each user.
name	VARCHAR(100)	NOT NULL	Full name of the user.
email	VARCHAR(100)	UNIQUE, NOT NULL	User's email address (must be unique).
password	VARCHAR(255)	NOT NULL	Encrypted/hashed password for authentication.
role	ENUM('admin','user')	Default: 'user'	Role assigned to the user (either admin or user).
profile_photo	VARCHAR(255)	NULL allowed	File path or URL of the user's profile photo.
currency	VARCHAR(10)	Default: 'INR'	Preferred currency of the user.
timezone	VARCHAR(50)	Default: 'Asia/Kolkata'	User's preferred timezone.
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Timestamp when the record was created.

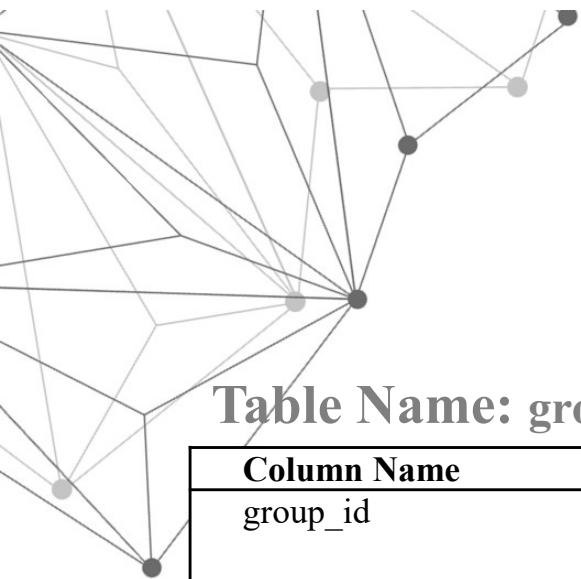
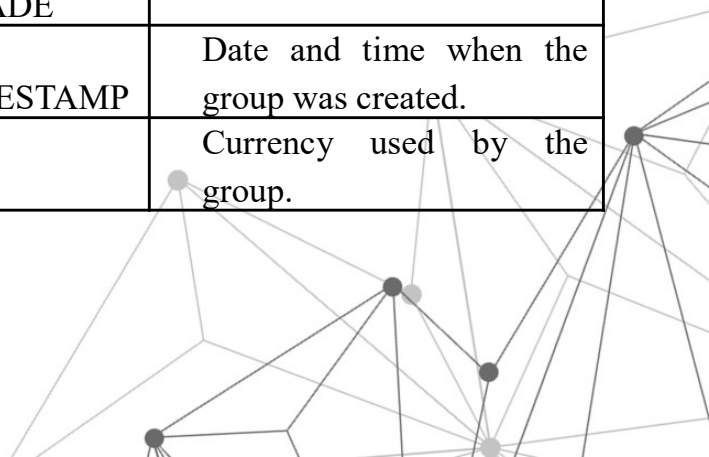


Table Name: groups

Column Name	Data Type	Constraints / Default	Description
group_id	INT (AUTO INCREMENT)	Primary Key , Auto-increment	Unique identifier for each group.
group_name	VARCHAR(100)	NOT NULL	Name of the group.
created_by	INT	NOT NULL, Foreign Key → users(user_id)ON DELETE CASCADE	ID of the user who created the group.
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Date and time when the group was created.
currency	VARCHAR(100)	NOT NULL	Currency used by the group.



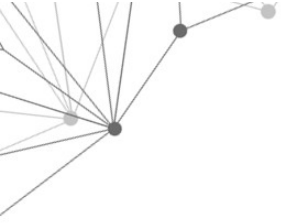


Table Name: group_members

Column Name	Data Type	Constraints / Default	Description
member_id	INT (AUTO_INCREMENT)	Primary Key , Auto-increment	Unique identifier for each membership record.
group_id	INT	NOT NULL, Foreign Key → groups(group_id) ON DELETE CASCADE	ID of the group the user is a member of.
user_id	INT	NOT NULL, Foreign Key → users(user_id) ON DELETE CASCADE	ID of the user who is a member of the group.
joined_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Date and time when the user joined the group.

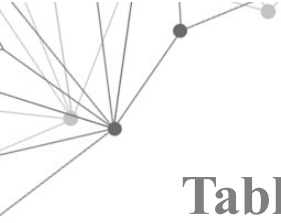


Table Name: expenses

Column Name	Data Type	Constraints / Default	Description
expense_id	INT (AUTO_INCREMENT)	Primary Key , Auto-increment	Unique identifier for each expense record.
group_id	INT	NOT NULL, Foreign Key → groups(group_id) ON DELETE CASCADE	ID of the group associated with the expense.
paid_by	INT	NOT NULL, Foreign Key → users(user_id) ON DELETE CASCADE	ID of the user who paid for the expense.
description	VARCHAR(255)	NOT NULL	Short text describing the expense.
category	VARCHAR(50)	NULL allowed	Category or type of the expense (e.g., food, travel).
amount	INT	NOT NULL	Total amount of the expense.
currency	VARCHAR(10)	Default: 'INR'	Currency in which the expense is recorded.
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Date and time when the expense was created.

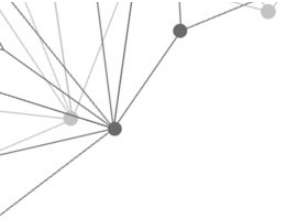


Table Name: expense_splits

Column Name	Data Type	Constraints / Default	Description
split_id	INT (AUTO_INCREMENT)	Primary Key , Auto-increment	Unique identifier for each split record.
expense_id	INT	NOT NULL, Foreign Key → expenses(expense_id) ON DELETE CASCADE	ID of the expense this split belongs to.
user_id	INT	NOT NULL, Foreign Key → users(user_id) ON DELETE CASCADE	ID of the user responsible for this share of the expense.
share_amount	INT	NOT NULL	Amount allocated to the user from the total expense.

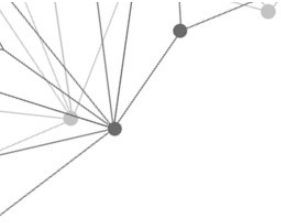


Table Name: settlements

Column Name	Data Type	Constraints / Default	Description
settlement_id	INT (AUTO_INCREMENT)	Primary Key , Auto-increment	Unique identifier for each settlement record.
group_id	INT	NOT NULL, Foreign Key → groups(group_id) ON DELETE CASCADE	ID of the group in which the settlement occurs.
from_user	INT	NOT NULL, Foreign Key → users(user_id)	ID of the user who pays the settlement amount.
to_user	INT	NOT NULL, Foreign Key → users(user_id)	ID of the user who receives the settlement amount.
amount	INT	NOT NULL	Amount of money being settled.
status	ENUM('pending','completed')	Default: 'pending'	Status of the settlement (either pending or completed).
settled_at	TIMESTAMP	NULL allowed	Date and time when the settlement was completed.

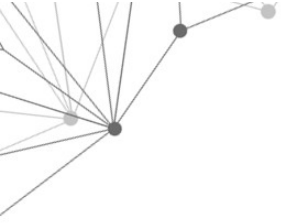


Table Name: notifications

Column Name	Data Type	Constraints / Default	Description
notification_id	INT (AUTO_INCREMENT)	Primary Key , Auto-increment	Unique identifier for each notification.
user_id	INT	NOT NULL, Foreign Key → users(user_id) ON DELETE CASCADE	ID of the user to whom the notification belongs.
message	TEXT	NOT NULL	The content or body of the notification.
is_read	BOOLEAN	Default: FALSE	Flag indicating whether the notification has been read (TRUE) or not (FALSE).
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Date and time when the notification was created.

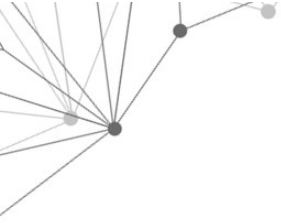
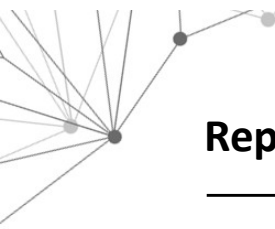


Table Name: queries (Admin Support)

Column Name	Data Type	Constraints / Default	Description
query_id	INT (AUTO_INCREMENT)	Primary Key , Auto-increment	Unique identifier for each support query.
user_id	INT	NOT NULL, Foreign Key → users(user_id) ON DELETE CASCADE	ID of the user who submitted the query.
category	ENUM('bug','payment','general')	Default: 'general'	Type of query: bug, payment, or general.
message	TEXT	NOT NULL	The content or details of the query.
status	ENUM('open','resolved')	Default: 'open'	Current status of the query (open or resolved).
created_at	TIMESTAMP	Default: CURRENT_TIMESTAMP	Date and time when the query was submitted.



Report Formats

01

Group-wise expense summary

02

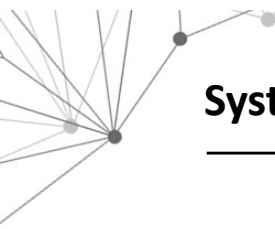
Individual user balance report

03

Settlement history

04

Query/Support report for admin



System Implementation

Module Implementation Details

User Module: Authentication (JWT), profile settings

Group Module: CRUD for groups, member invitations

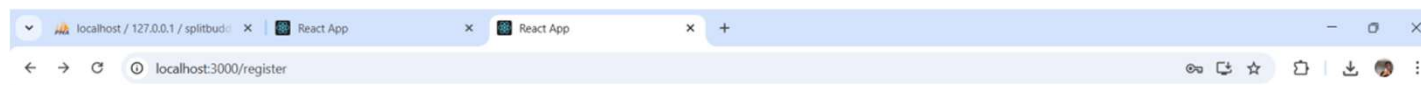
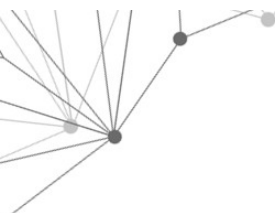
Expense Module: Add/view expenses, split logic

Settlement Module: Track and mark settlements as completed

Notifications: Real-time messaging and alerts

Admin Module: Monitoring groups and User management

- Manage queries and bug reports





Create Your Account

Full Name

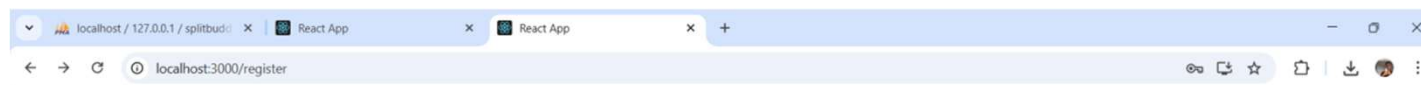
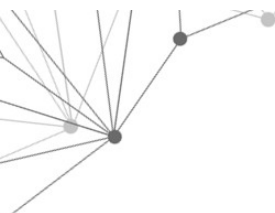
Email

Password

[Create Account](#)

Already have an account?... [Login](#)





Create Your Account

Registration successful. Please login.

Full Name

ishaSoni

Email

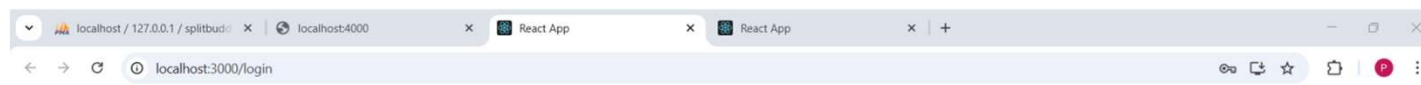
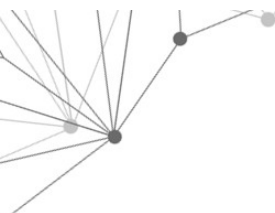
ishaa27@gmil.com

Password

Create Account

Already have an account?... [Login](#)







Welcome Back!

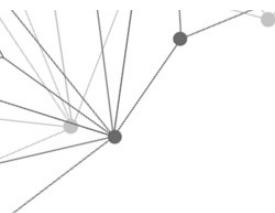
Email

Password

Login

New user?... [Register here](#)
[Forgot Password ??](#)





localhost / 127.0.0.1 / splitbud...localhost:4000React AppReact App

localhost:3000/user/home

SplitBuddy

HomeGroupsBalancesContact UsSettings

Welcome, virat kohliLogout

SplitBuddy helps you manage group expenses in real time, with smart tracking and instant updates for every member.

Groups Joined

4

Total groups you are part of

Expenses Participated

11

Total expenses you are part of

Create New Group

Currency Converter

1us USDtoIN INRConvert

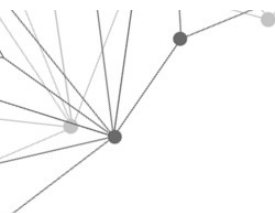
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27°C Sunny

Search

ENG IN

10:53 15-10-2025



localhost / 127.0.0.1 / splitbud x localhost:4000 x React App x React App x +

localhost:3000/groups

Welcome, virat kohli Logout

localhost:3000 says
Group created successfully!

OK

Create New Group

GOA TRIP INR ₹

GROUP'S

GOA TRIP (ID: #51)
₹ INR

DETAIL'S

+ Add Members

Expenses Delete Edit

1 - Isha soni (isha@gmail.com) (ADMIN)
2 - Neeludhanawla (neel@gmail.com)
3 - Vruti (vruti@gmail.com)
4 - Nandani chauhan (nandani@gmail.com)
5 - virat kohli (24melt01@charusat.edu.in)
6 - Nidhi Patel (splitbuddy2025@gmail.com)
7 - Manan Karsara (24melt01@charusat.edu.in)

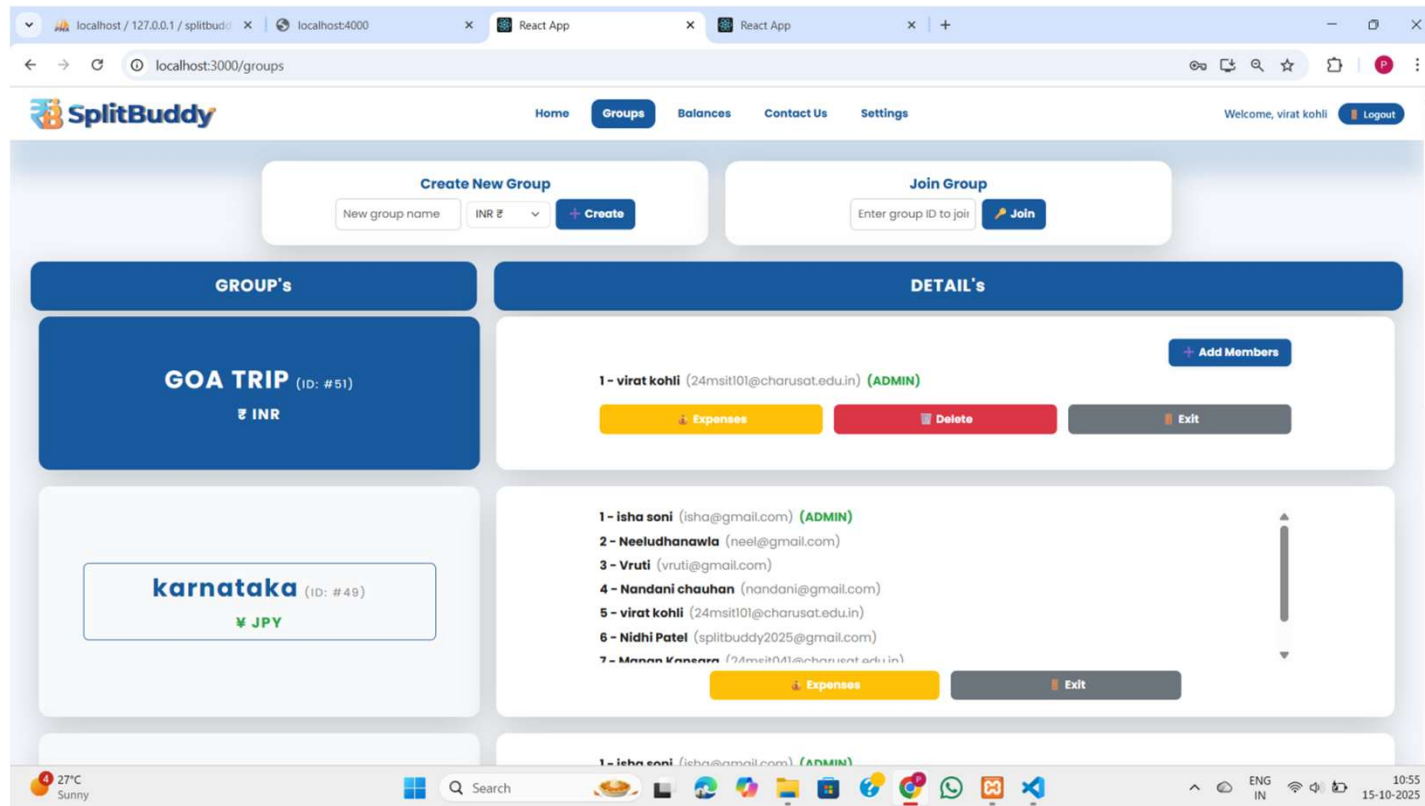
Expenses Edit

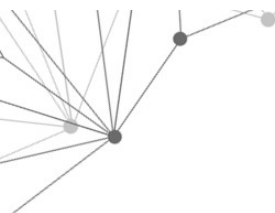
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Search

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localhost:3000/groups

HomeGroupsBalancesContact UsSettings

Welcome, virat kohliLogout

Create New Group

Join Group

GROUP's

GOA TRIP (ID: #51)
₹ INR

karnataka (ID: #49)
¥ JPY

DETAIL's

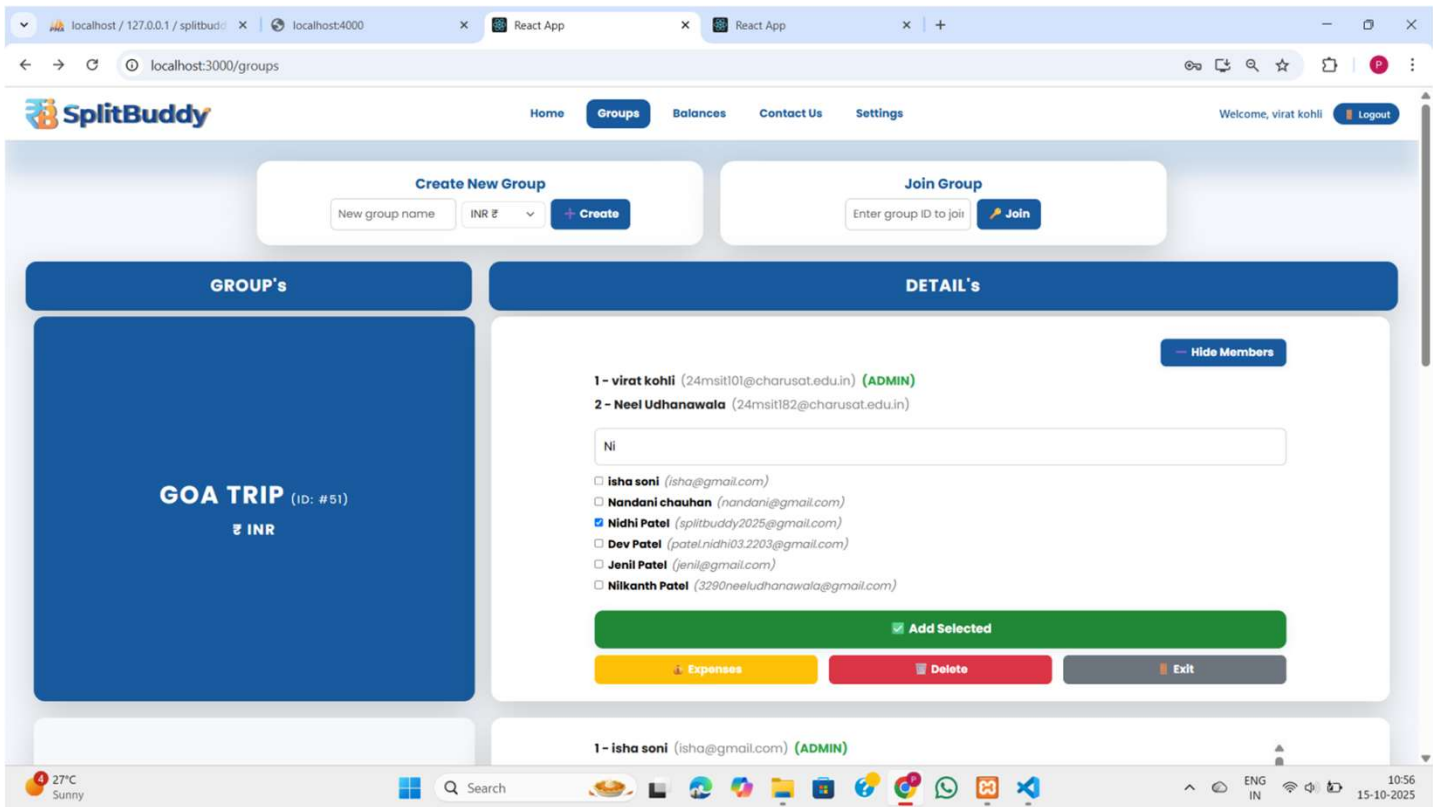
1 - virat kohli (24msit101@charusat.edu.in) (ADMIN)
ExpensesDeleteExit

1 - isha soni (isha@gmail.com) (ADMIN)
2 - Neeludhanawla (neel@gmail.com)
3 - Vruti (vruti@gmail.com)
4 - Nandani chauhan (nandani@gmail.com)
5 - virat kohli (24msit101@charusat.edu.in)
6 - Nidhi Patel (splitbuddy2025@gmail.com)
7 - Manan Kansara (2Ameit041@charusat.edu.in)
ExpensesExit

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Search

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localhost / 127.0.0.1 / splitbud...localhost:4000React AppReact App

localhost:3000/expenses/51

SplitBuddyHomeGroupsBalancesContact UsSettingsWelcome, virat kohliLogout

GOA TRIP - Expenses

+ Add Expense

Description

Amount

Paid By:

-- Select Payer --

Split Between:

☐ Neel Udhanawala

☐ Nidhi Patel

☐ Nandani chauhan

+ Add Expense

Expenses

Expense added successfully!

Split Details:

• virat kohli has to pay ₹1250

• Neel Udhanawala has to pay ₹1250

• Nidhi Patel has to pay ₹1250

• Nandani chauhan has to pay ₹1250

virat kohli paid ₹5000 for Dinner (10/15/2025)

Split between:

virat kohli ₹1250

Neel Udhanawala ₹1250

Nidhi Patel ₹1250

Nandani chauhan ₹1250

Generate Final Settlement

Final Settlements

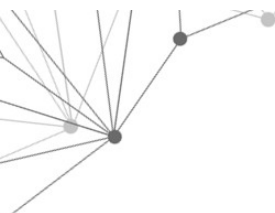
All settlements are paid!

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localhost / 127.0.0.1 / splitbud...localhost:4000React AppReact App

localhost:3000/expenses/51

SplitBuddyHomeGroupsBalancesContact UsSettingsWelcome, virat kohliLogout

GOA TRIP - Expenses

+ Add Expense

Description

Amount

Paid By:
-- Select Payer --

Split Between:

☐ virat kohli

☐ Neel Udhanawala

☐ Nidhi Patel

+ Add Expense

Expenses

Expense added successfully!

Split Details:

• Neel Udhanawala has to pay ₹333

• Nidhi Patel has to pay ₹333

• virat kohli has to pay ₹333

Neel Udhanawala paid ₹1000 for Rides (10/15/2025)

Split between:
Neel Udhanawala ₹333
Nidhi Patel ₹333
virat kohli ₹333

virat kohli paid ₹5000 for Dinner (10/16/2025)

Split between:
virat kohli ₹1250
Neel Udhanawala ₹1250

Generate Final Settlement

Final Settlements

All settlements are paid!

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Search

ENG IN10:58 15-10-2025

localhost / 127.0.0.1 / splitbud...localhost:4000React AppReact App

localhost:3000/expenses/51

☐ Nidhi Patel

+ Add Expense

virat kohli ₹333

virat kohli paid ₹5000 for Dinner (10/15/2025)

Split between:

virat kohli ₹1250

Neel Uddhanawala ₹1250

Nidhi Patel ₹1250

Nandani chauhan ₹1250

Generate Final Settlement

Final Settlements

Nandani chauhan pays ₹1250 to virat kohli pending

virat kohli pays ₹15417 to Nidhi Patel pendingPay

Neel Uddhanawala pays ₹917 to virat kohli pending

Neel Uddhanawala pays ₹16334 to Nidhi Patel pending

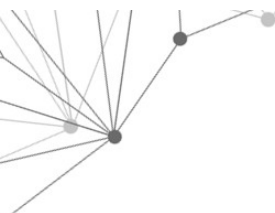
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localhost / 127.0.0.1 / splitbud...localhost:4000React AppReact App

localhost:3000/settle-up

SplitBuddyHomeGroupsBalancesContact UsSettingsWelcome, virat kohliLogout

Settle Up

Select User

Nidhi Patel

Amount

15417

Choose Payment Method:

Pay on HandPay by Razorpay

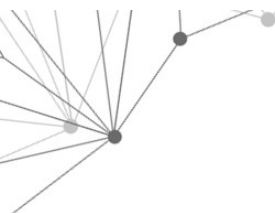
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localhost / 127.0.0.1 / splitbud...localhost:4000React AppReact App

localhost:3000/settle-up

Test Mode

S

SplitBuddy

Price Summary

₹15,417

Using as +91 87581 76552

You will be redirected in 4 seconds

Payment Successful

✓

SplitBuddy

₹15,417

Oct 15, 2025, 10:59 AM

Netbanking

pay_RTdLVdwt1a6zF6

Visit razorpay.com/support for queries

Secured by

Razorpay

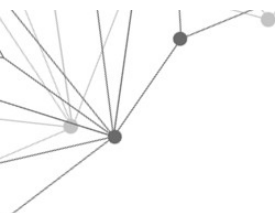
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27°C Sunny

Search

ENG IN

10:59 15-10-2025



localhost / 127.0.0.1 / splitbud...localhost:4000React AppReact App

localhost:3000/settle-up

SplitBuddyHomeGroupsBalancesContact UsSettingsWelcome, virat kohliLogout

Settle Up

Select User

Nilkanth Patel

Amount

5000

Choose Payment Method:

Pay on Hand

Pay by Razorpay

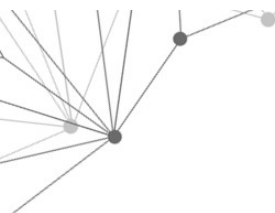
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ENG IN

11:29 15-10-2025



localhost / 127.0.0.1 / splitbud

localhost:4000

React App

React App

localhost:3000/balances

Your Balances

You are owed in Sarangpur

₹ 15384397

[Show Expenses](#)

You are owed in Ladakh

₹ 3452

[Show Expenses](#)

You owe in holiday

₹ 1158

[Show Expenses](#)

You owe in karnataka

₹ 839

[Show Expenses](#)

You owe in GOA TRIP

₹ 13250

[Hide Expenses](#)

Description	Amount	Currency	Paid By	Date
Lunch	₹50000	INR	Nidhi Patel	10/15/2025
Rides	₹1000	INR	Neel Udhanawala	10/15/2025
Dinner	₹5000	INR	virat kohli	10/15/2025

Detailed Pending Settlements

Manan owes you ₹5366674 in Sarangpur

Neeludhanawla owes you ₹5337634 in Sarangpur

Neel Udhanawala owes you ₹4680088 in Sarangpur

You owe krish shah ₹1158 in holiday

You owe isha soni ₹839 in karnataka

Nandani chauhan owes you ₹1250 in GOA TRIP

Neel Udhanawala owes you ₹917 in GOA TRIP

27°C Sunny

Search

ENG IN

11:00 15-10-2025

localhost / 127.0.0.1 / splitbud

localhost:4000

React App

React App

localhost:3000/balances

Your Balances

You are owed in Sarangpur

₹ 15384397

Show Expenses

You are owed in Ladakh

₹ 3452

Show Expenses

You owe in holiday

₹ 1158

Show Expenses

You owe in karnataka

₹ 839

Show Expenses

You owe in GOA TRIP

₹ 13250

Hide Expenses

Description	Amount	Currency	Paid By	Date
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Detailed Pending Settlements

Manan owes you ₹5366674 in Sarangpur

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You owe isha soni ₹839 in karnataka

Nandani chauhan owes you ₹1250 in GOA TRIP

Neel Udhanawala owes you ₹917 in GOA TRIP

GOA TRIP_statement.csv

515 B • Done

Received Files

GOA TRIP_statement.csv

C:\Users\patel\Downloads

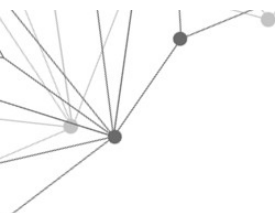
Open

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Search

ENG IN

11:00 15-10-2025



W Home

GOA TRIP_statement.csv

New

Upgrade now

File Edit View Insert Page Layout Formulas Data Review View Tools Smart Toolbox

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A1 fx Group: GOA TRIP

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	Group: GOA TRIP																			
2																				
3	Description	Amount	Paid By	Date	Split Between															
4	Lunch	50000	Nidhi Patel	2025-10-15	virat kohli (INR 16667); Neel Udhanawala (INR 16667); Nidhi Patel (INR 16667)															
5	Rides	1000	Neel Udhan	2025-10-15	Neel Udhanawala (INR 333); Nidhi Patel (INR 333); virat kohli (INR 333)															
6	Dinner	5000	virat kohli	2025-10-15	virat kohli (INR 1250); Neel Udhanawala (INR 1250); Nidhi Patel (INR 1250); Nandani chauhan (INR 1250)															
7																				
8	Total Balance	-13250																		
9																				
10																				
11																				
12																				
13																				
14																				
15																				
16																				
17																				
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25																				
26																				

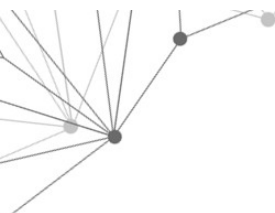
GOA TRIP_statement

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Search

ENG IN

11:01 15-10-2025



localhost / 127.0.0.1 / splitbud... x localhost:4000 x React App x React App x +

localhost:3000/contactus

🔍 📄 ⚙️ ⌵



Home Groups Balances **Contact Us** Settings

Welcome, virat kohli **Logout**

 **Contact Us**

✔️ Your query has been submitted successfully!

Name

Email

Category

Message

Send Message

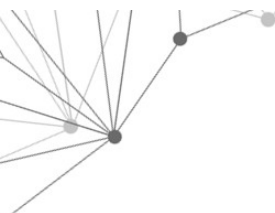
🌤️ 27°C Sunny





ENG IN

11:01 15-10-2025



localhost / 127.0.0.1 / splitbud...localhost:4000React AppReact App

localhost:3000/settings

SplitBuddyHomeGroupsBalancesContact UsSettingsWelcome, virat kohliLogout

Settings

Name

virat kohli

Email

24msit01@charusat.edu.in

New Password

Leave blank to keep current password

Save Changes

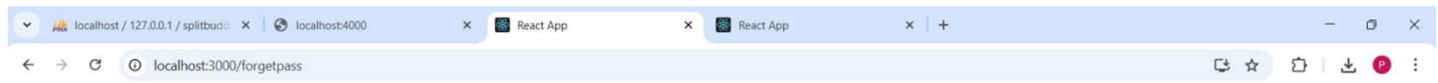
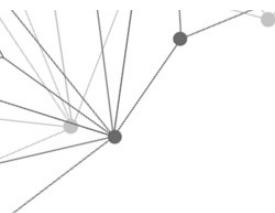
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Search

ENG IN

11:02 15-10-2025

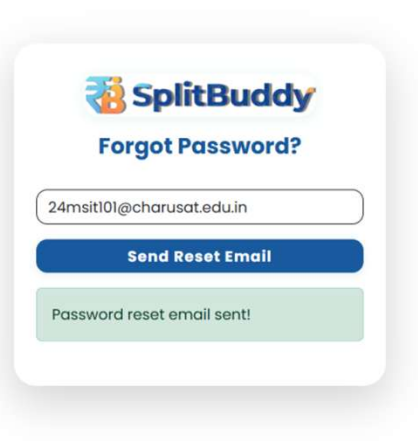
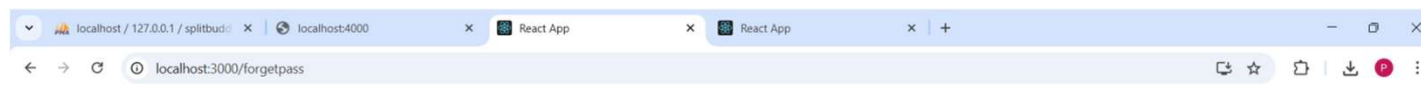
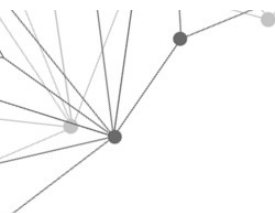


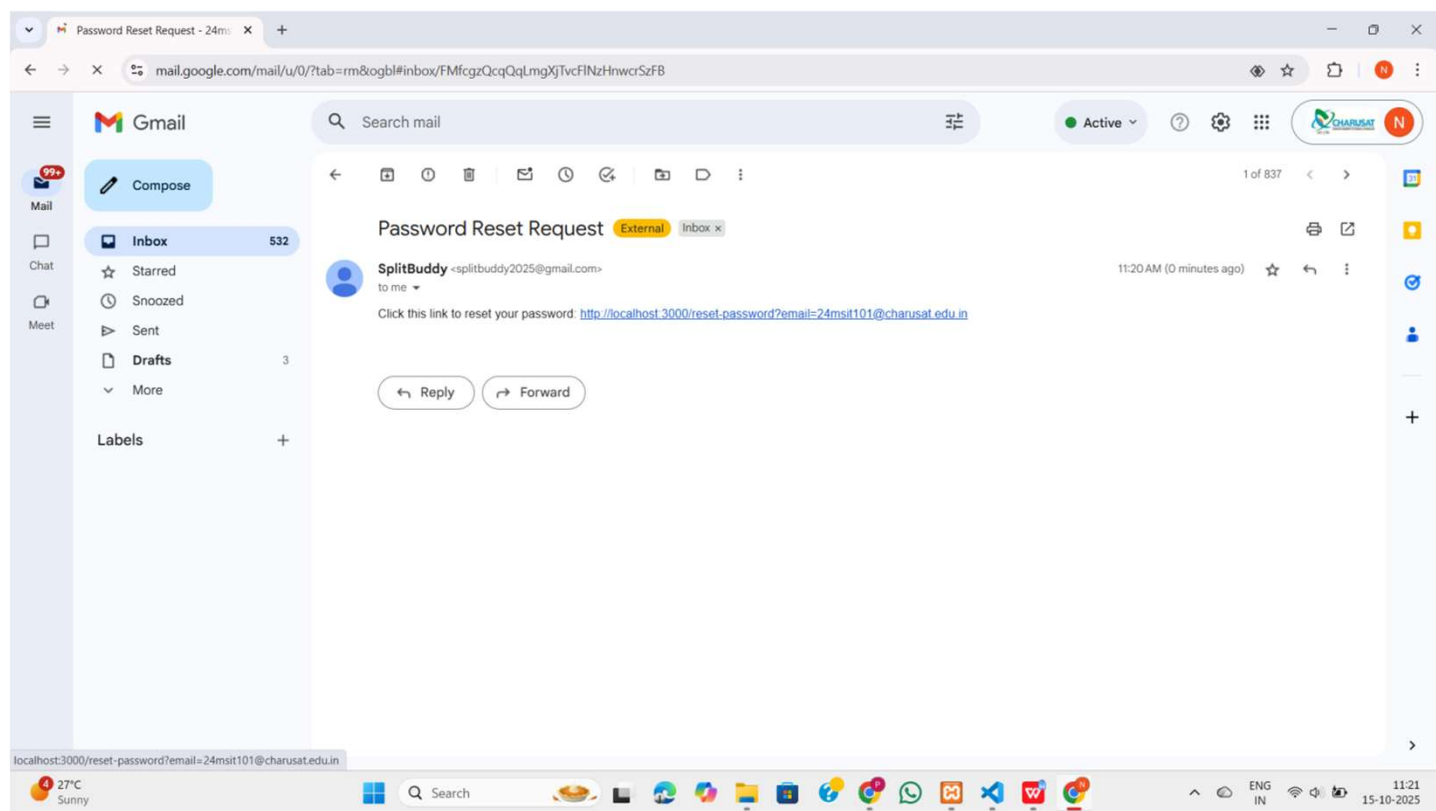


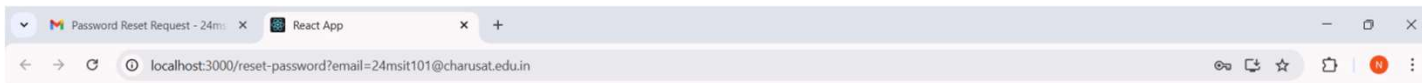
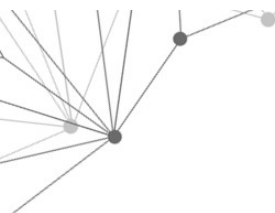
Forgot Password?

Send Reset Email







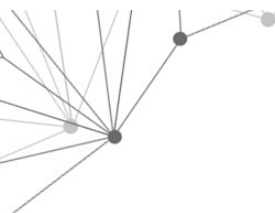




Reset Password

Reset Password





Create Your Account

Full Name

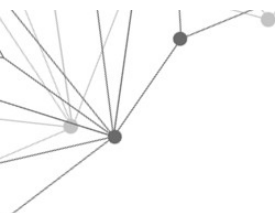
Email

Password

Create Account

Already have an account?... [Login](#)







Welcome Back!

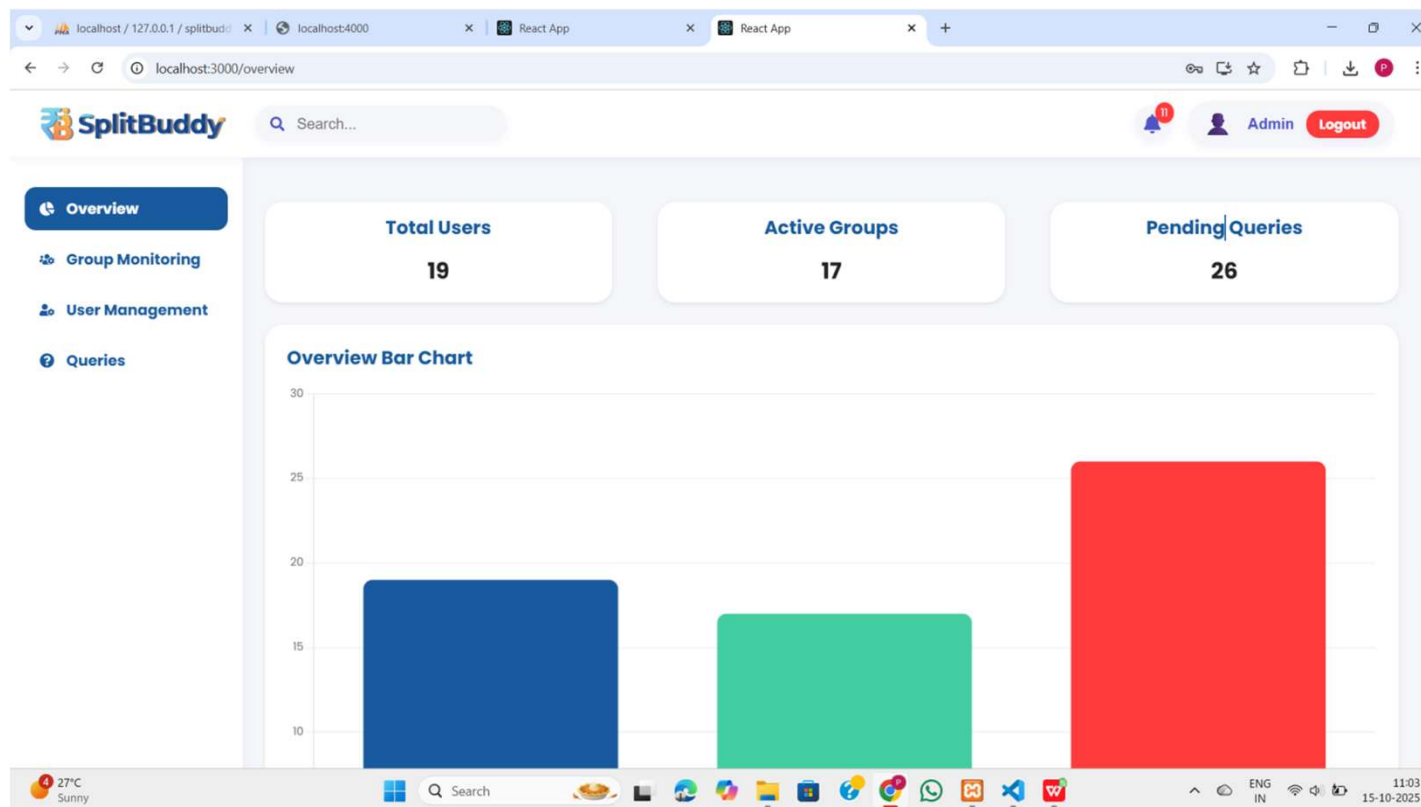
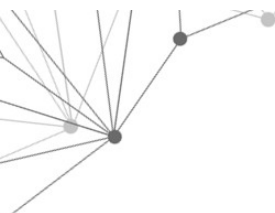
Email

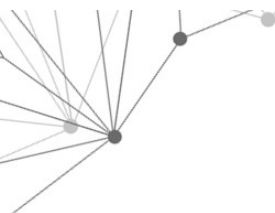
Password

Login

New user?... [Register here](#)







localhost / 127.0.0.1 / splitbud...

localhost:4000

React App

React App

localhost:3000/overview

SplitBuddy

Search...

Admin

Logout

Overview

Group Monitoring

User Management

Queries

Total Users

19

Active Groups

17

Overview Bar Chart

Metric	Value
Total Users	19
Active Groups	17

Notifications

New group created: GOA TRIP by virat kohli

New group created: New York by isha soni

New group created: karnataka by isha soni

New group created: kashmir by isha soni

New group created: holiday by isha soni

New group created: jaipurrr by isha soni

New group created: thailanddd by isha soni

New group created: Ladakh by virat kohli

New group created: sarangpur by Nilkanth Patel

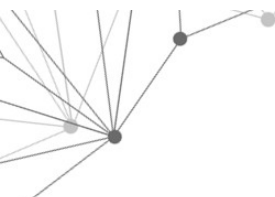
Welcome Nilkanth Patel!!

27°C Sunny

Search

ENG IN

11:03 15-10-2025



localhost / 127.0.0.1 / splitbud...localhost:4000React AppReact App

localhost:3000/groupmonitoring

SplitBuddySearch...AdminLogout

Overview

Group Monitoring

User Management

Queries

GOA TRIP

Members: 6

V

virat kohli

24msit101@charusat.edu.in

N

Neel Udhanawala

24msit182@charusat.edu.in

N

Nidhi Patel

splitbuddy2025@gmail.com

N

Nandani chauhan

nandani@gmail.com

N

Nilkanth Patel

3290neeludhanawala@gmail.com

I

isha soni

isha@gmail.com

New York

Members: 4

I

isha soni

isha@gmail.com

N

Neel Udhanawala

24msit182@charusat.edu.in

N

Nilkanth Patel

3290neeludhanawala@gmail.com

N

Nidhi Patel

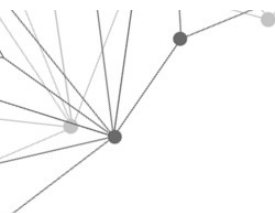
splitbuddy2025@gmail.com

27°C Sunny

Search

ENG IN

11:03 15-10-2025



React App

localhost / 127.0.0.1 / splitbuddy

localhost:3000/groupmonitoring

Search pizza

Admin [Logout](#)

Overview

Group Monitoring

User Management

Queries

pizza party

Members: 3

bhavya shah
bhavya@gmail.com

Abhi
abhi@123gmail.com

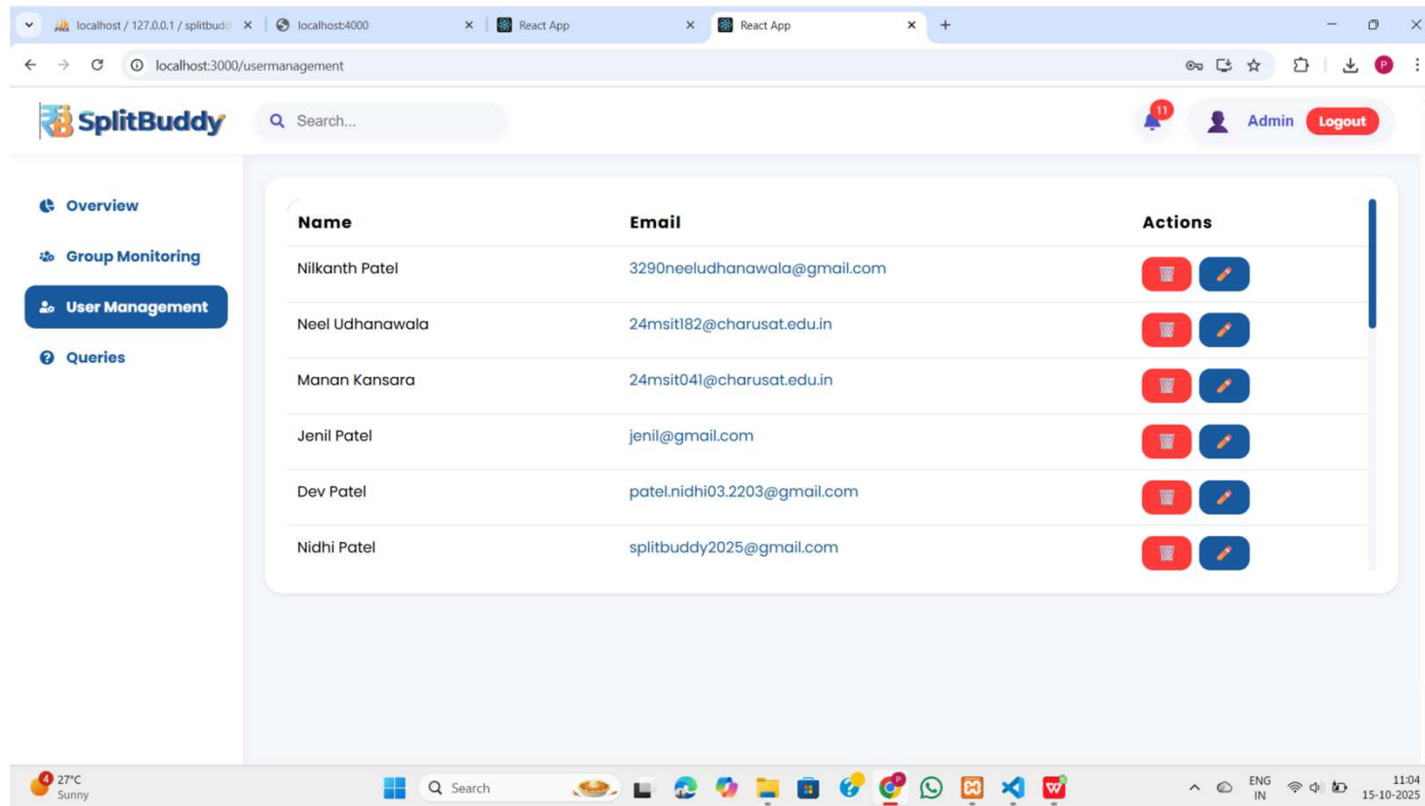
Manan
manan@gmail.com

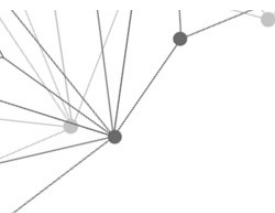
9:27°C
H17%

Search

ENG
IN

62-22
6/20/2025





React App

localhost / 127.0.0.1 / splitbuddy

localhost:3000/usermanagement

 SplitBuddy

neel

 0

 Admin

Logout

Overview

Group Monitoring

User Management

Queries

Name	Email	Actions
Neel	neel@gmail.com	DeleteEdit

Edit User

Name

Neeludhanawid

Email

neel@gmail.com

Save

Cancel

27°C

Here

Search



ENG

IN

02:22

20/09/2025

React App

localhost / 127.0.0.1 / splitbuddy

localhost:3000/usermanagement

 Search...

 0

 Admin

[Logout](#)

Overview

Group Monitoring

User Management

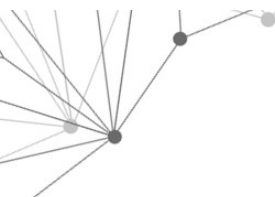
Queries

Name	Email	Actions
Manan	manan@gmail.com	Delete Edit
Abhi	abhi@123gmail.com	Delete Edit
Neeludhanawla	neel@gmail.com	Delete Edit
Vruti	vruti@gmail.com	Delete Edit
sonal patel	sonal@gmail.com	Delete Edit
isha soni	isha@gmail.com	Delete Edit

9:27°C
Hydr



END
TN 02:23
20/10/2023



React App

localhost / 127.0.0.1 / splitbuddy

+

localhost:3000/usermanagement

 SplitBuddy

Search...

Overview

Group Monitoring

User Management

Queries

localhost:3000 says

Are you sure you want to remove this user?

OK Cancel

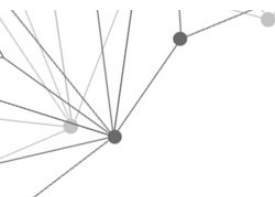
Admin Logout

Name	Email	Actions
Manan	manan@gmail.com	Delete Edit
Abhi	abhi@123gmail.com	Delete Edit
Neeludhanawla	neel@gmail.com	Delete Edit
Vruti	vruti@gmail.com	Delete Edit
sonal patel	sonal@gmail.com	Delete Edit
isha soni	isha@gmail.com	Delete Edit

9:27°C Home

Search

ENG IN 02:23 20/09/2025



React App

localhost / 127.0.0.1 / splitbuddy

localhost:3000/usermanagement

localhost:3000 says
Can not remove user because user is in a group.
OK

SplitBuddy

Search...

Overview

Group Monitoring

User Management

Queries

Name	Email	Actions
Manan	manan@gmail.com	Delete Edit
Abhi	abhi@123gmail.com	Delete Edit
Neeludhanawla	neel@gmail.com	Delete Edit
Vruti	vruti@gmail.com	Delete Edit
sonal patel	sonal@gmail.com	Delete Edit
isha soni	isha@gmail.com	Delete Edit

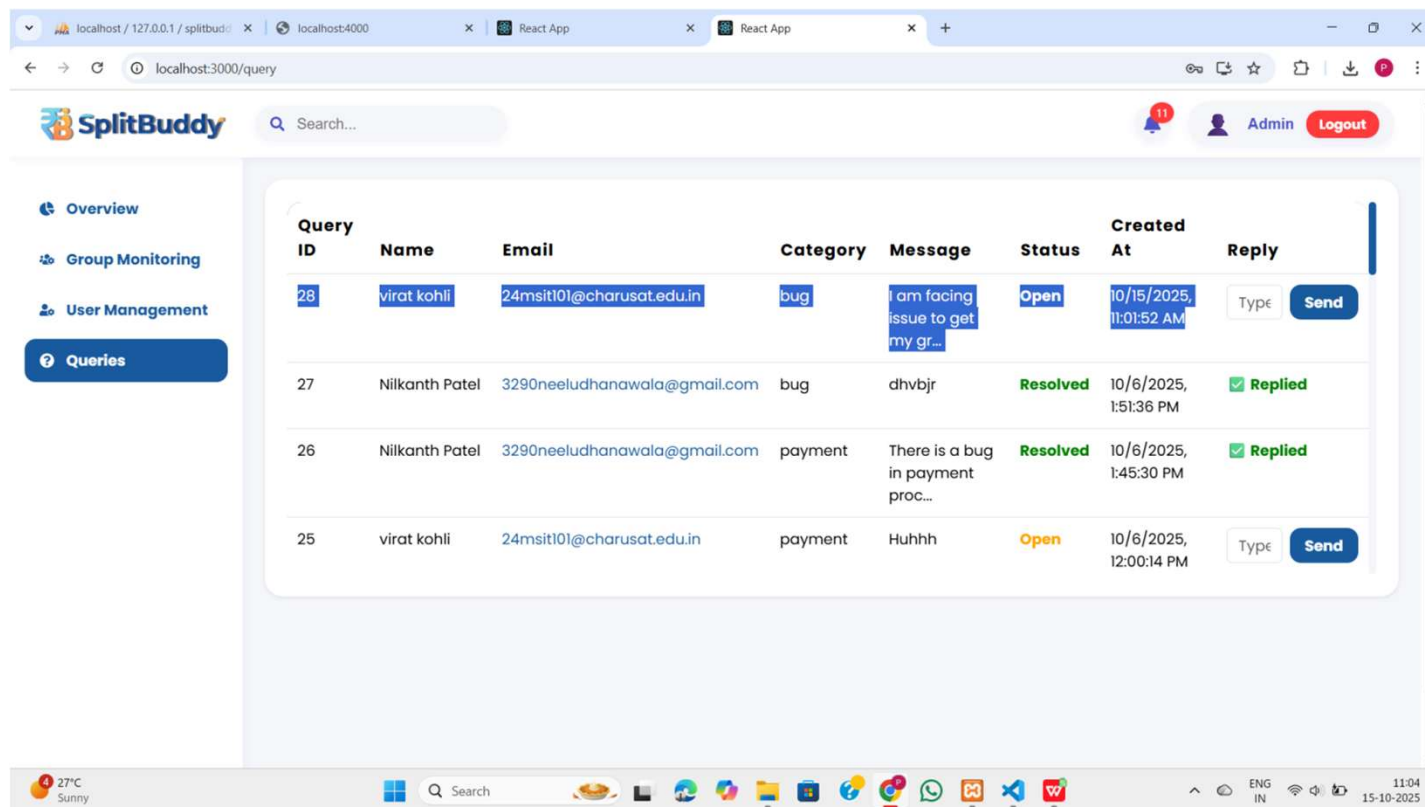
Admin Logout

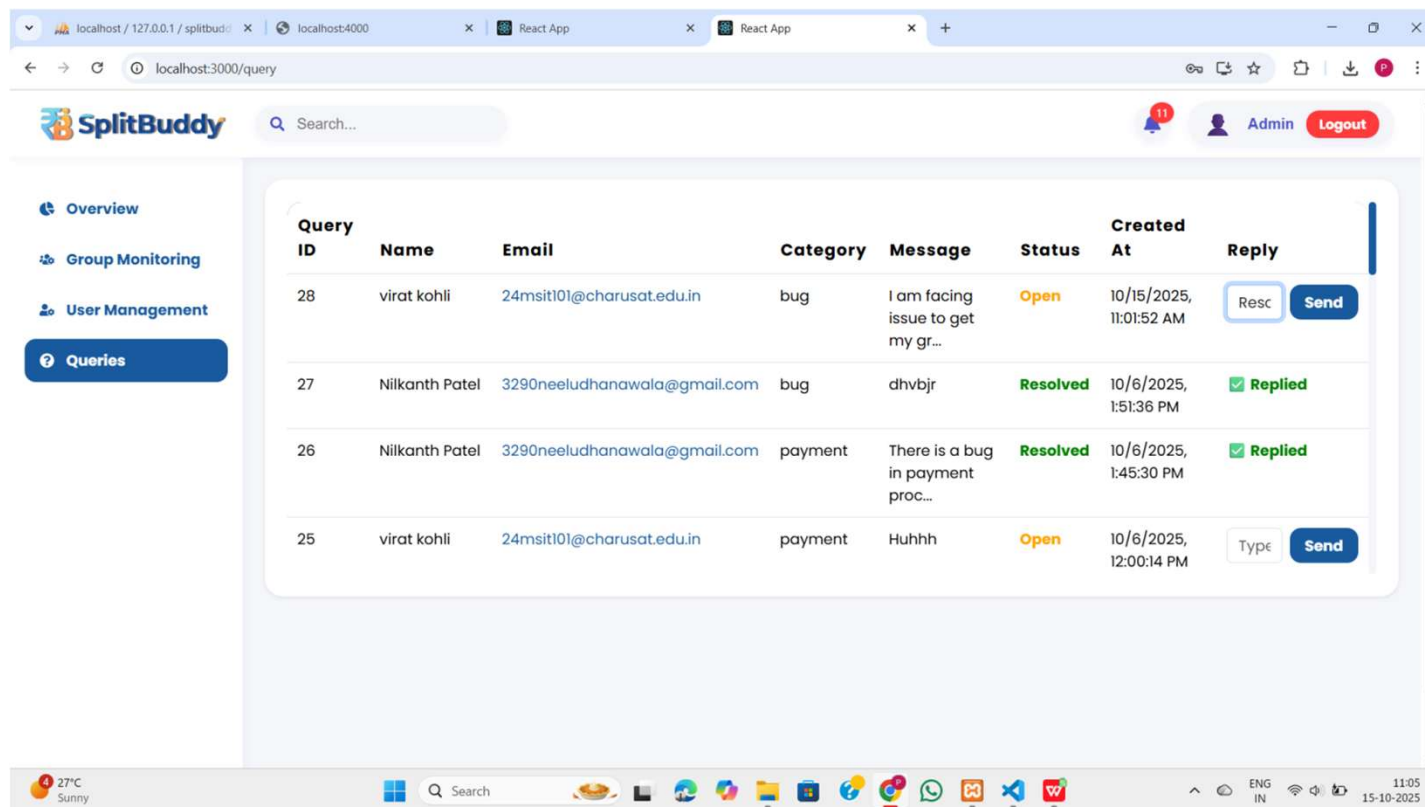
27°C Haze

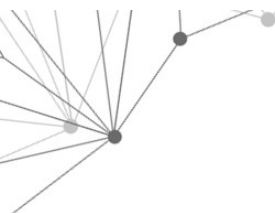
Search

46

ENG IN 02:23 20/09/2025







Your Query Has Been Resolved | x +

mail.google.com/mail/u/0/?tab=rm&ogbl#inbox/FMfcgzQcQqLmfQcpKSwrXKXvbWKXlCW

☰

Gmail

Compose

Inbox 532

Starred

Snoozed

Sent

Drafts 3

More

Labels +

Search mail

Active

CHARUSAT

←

📄

🕒

🗑️

✉️

🕒

🔄

📁

🗑️

⋮

1 of 836

📄

📄

📄

Your Query Has Been Resolved ✓ External Inbox x

SplitBuddy Support <splitbuddy2025@gmail.com>

to me

11:05 AM (0 minutes ago) ☆ ↶ ⋮

Query Resolved

Hi,

Your query has been resolved. Here's the response from our team:

Resolved

Thank you for contacting SplitBuddy!

↶ Reply

↷ Forward

27°C Sunny

🪟

🔍 Search

🍲

🖨️

🌐

🎨

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📧

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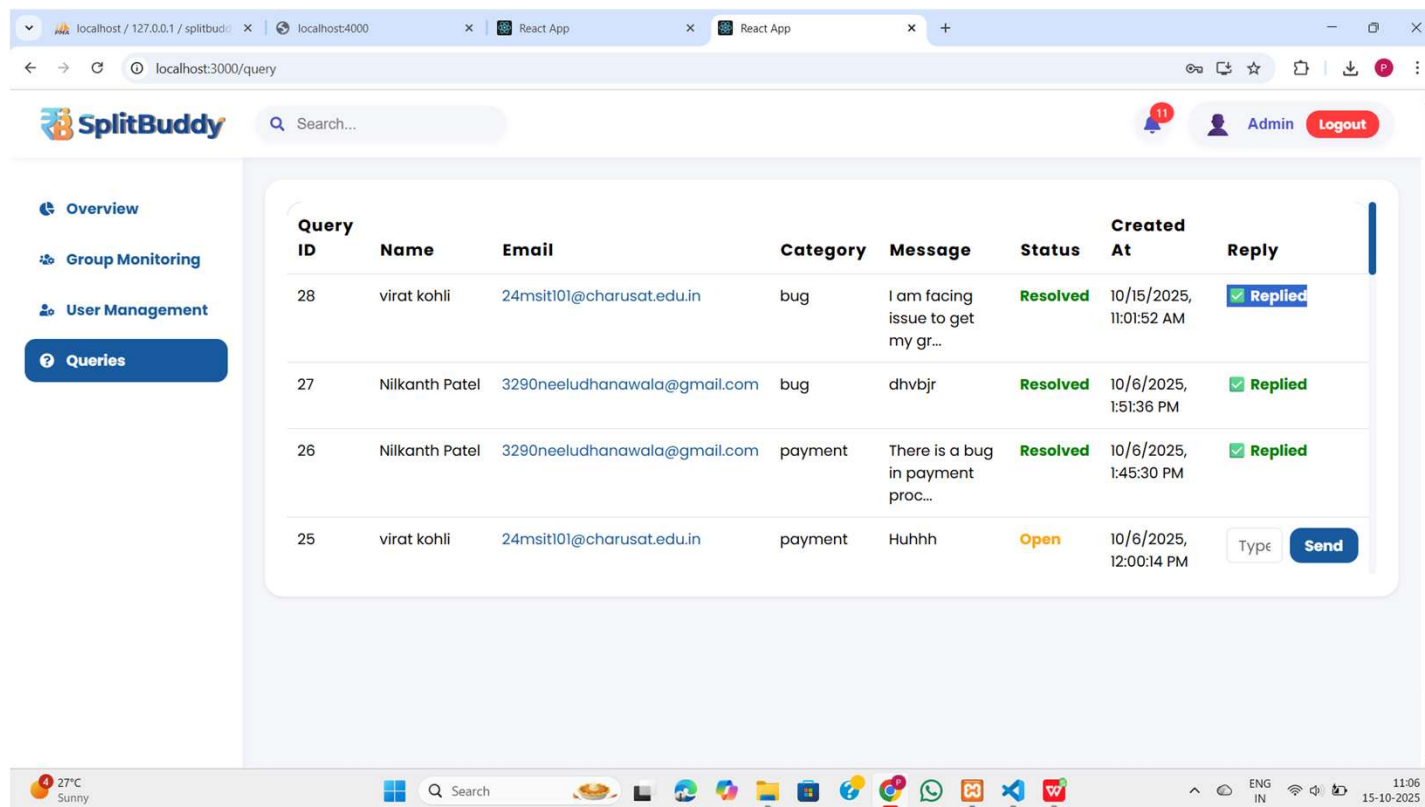
ENG IN

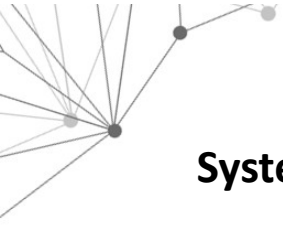
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🔋

11:05 15-10-2025





System Testing

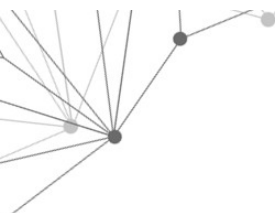
Testing Strategies

Unit Testing: Validate individual modules (login, expense split).

Integration Testing: Ensure modules work together (expense + settlement).

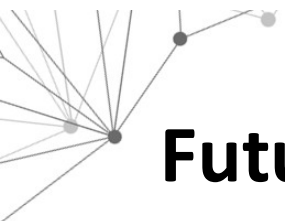
System Testing: Verify overall functionality.

User Acceptance Testing: Feedback from sample users.



Test Cases & Results

Test ID	Case Description	Input	Expected Output	Results
TC01	User login with valid credentials	Correct email & password	Redirect to dashboard	Pass
TC02	User login with invalid credentials	Wrong password	Error message: "Invalid username or password"	Pass
TC03	Add expense (normal)	Amount = 500, payer = User A, members = A,B,C	Expense saved and split calculated	Pass
TC04	Add expense with empty fields	Blank amount field	Validation error shown	Pass
TC05	Generate Settlement	Button Click	Show generated settlement between members	Pass
TC06	Settlement update	Pay amount = 200	Status updated to "Completed"	Pass



Future Enhancement

01

Mobile app
(Android/iOS)

02

Payment gateway
integration (UPI, PayPal)

03

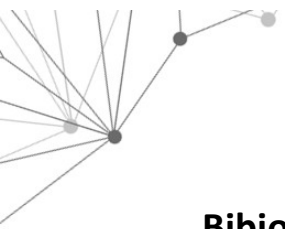
Offline mode with
data sync

04

🔍AI-powered expense
🔍categorization

05

Multi-language
support



Bibliography / References

01

React Documentation – <https://react.dev>

02

Node.js Docs – <https://nodejs.org>

03

MySQL Reference Manual – <https://dev.mysql.com/doc>

04

Bootstrap Docs – <https://getbootstrap.com>

THANK YOU

